Interatomic potentials for molten alkali chlorides and their mixtures

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Abstract

Molten alkali chlorides are key materials for advanced clean energy technologies due to their high thermal stability and favorable thermophysical properties. However, accurate prediction of their behavior, particularly in multicomponent mixtures at high temperatures, remains challenging. In this work, we develop a moment tensor potential (MTP) for molten NaCl, KCl, their binary mixtures (NaCl–KCl), and the NaK system, trained on an extensive quantum-mechanical database to ensure transferability across thermodynamic conditions. Large-scale molecular dynamics simulations with D3 dispersion correction are used to compute densities, diffusion coefficients, and structural properties over wide temperature ranges. The predicted NaCl–KCl phase diagram agrees well with experimental data, demonstrating the robustness of the model. The proposed framework provides an efficient and accurate tool for large-scale simulations of molten alkali chlorides and supports future studies of complex molten salt systems for clean energy applications.

I. INTRODUCTION

The development of advanced clean energy technologies has accelerated in response to the growing demand for sustainable, efficient, and low-carbon energy systems[1]. Applications such as large-scale thermal energy storage, concentrated solar power plants, and next-generation nuclear reactors require materials capable of operating reliably under extreme thermal and chemical conditions[2,3]. Within this context, molten salts have emerged as attractive candidates owing to their favorable physicochemical properties and high-temperature stability[4]. Molten salts typically exhibit wide liquid-phase temperature windows, low vapor pressure, high heat capacity, moderate thermal conductivity, and relatively low viscosity, making them well suited for use as heat-transfer fluids, coolants, and thermal energy storage media[5,6]. Their chemical stability and compositional flexibility further enable tuning of properties to meet specific operational requirements. A detailed understanding of their structural, thermodynamic, and transport properties is therefore essential for the rational design and optimization of molten salt–based energy systems[7].

Molten salts are commonly used in practical applications as multicomponent mixtures, where interactions among different ionic species can significantly influence macroscopic properties[8]. Accurate characterization of both pure salts and their mixtures is therefore of fundamental importance. In this context, phase diagram analysis plays a central role in determining phase stability, melting behavior, and compositional limits as functions of temperature and pressure[9,10]. The NaCl–KCl system, in particular, provides critical insight into the equilibrium between solid and liquid phases over a wide range of compositions, temperatures, and pressures. Precise knowledge of this phase diagram is essential for materials design and molten salt processing, as well as for high-temperature applications such as metallurgy, thermal energy storage, and nuclear reactor technologies. Reliable prediction of phase equilibria is therefore key to selecting optimal compositions and ensuring long-term operational stability in these systems.

Extensive experimental[11–15] and theoretical studies[7,16–18] have been conducted to characterize molten salt systems. Experimental techniques—including calorimetry, X-ray and neutron diffraction, and electrochemical measurements—have provided critical insights into the structural, thermodynamic, and transport properties of molten salt system[11,12,19–21].Nevertheless, experimental studies remain inherently limited by the harsh experimental conditions associated with molten salts—such as elevated temperatures, corrosive environments, and stringent safety requirements—which hinder comprehensive exploration across wide composition, temperature, and pressure ranges[22,23]. Moreover, the identification of optimal salt compositions within vast compositional spaces is highly challenging because accurate determination of key thermophysical and transport properties requires extensive, time-intensive, and costly high-temperature experiments. These challenges are further exacerbated in multicomponent systems, where the combinatorial growth of possible compositions makes purely experimental approaches impractical for systematic exploration[7].

These limitations motivate the use of theoretical and computational approaches capable of describing molten salts at the atomic scale. Atomistic modeling and molecular dynamics (MD) simulations provide a powerful framework for investigating structural organization, thermodynamic behavior, and transport phenomena under

controlled conditions. Among first-principles methods, density functional theory (DFT)[24] offers an accurate description of electronic structure, bonding, and charge redistribution, enabling reliable prediction of structural and dynamical properties[25,26]. However, the high computational cost of DFT restricts accessible system sizes and simulation times, limiting its applicability to short time scales and relatively small cells, and hindering the investigation of long-range correlations, slow transport processes, and phase behavior under realistic thermodynamic conditions.

To overcome these limitations, various semi-empirical interatomic potentials have been developed, including rigid-ion[27] and polarizable-ion models[28] as well as variable-charge force fields such as ReaxFF[29] and COMB[30].While these approaches substantially reduce computational cost and enable larger-scale simulations, they rely on predefined functional forms and system-specific parameterizations. Consequently, their transferability across different compositions and thermodynamic conditions is often limited, and extension to new chemical systems typically requires extensive reparameterization, which may compromise predictive accuracy[31–34].

The broad compositional and thermodynamic landscape of molten salt systems therefore calls for modeling strategies that are both accurate and transferable. Machine-learning interatomic potentials[35] have demonstrated the ability to achieve near first-principles accuracy while retaining computational efficiency comparable to classical force fields by learning complex many-body interactions directly from high-level reference data. Nevertheless, reliable interatomic potentials capable of jointly and accurately describing the liquid phase of technologically relevant molten salts—such as NaCl, KCl, NaCl–KCl mixtures, and NaK systems—over wide compositional and temperature ranges remain scarce. Previous neural-network interatomic potentials have been successfully applied to model molten NaCl, achieving predictive accuracy comparable to experimental data[7]. Key thermophysical properties, including the equation of state, density, heat capacity, diffusion coefficient, thermal conductivity, and melting point, were reproduced with high fidelity. In addition, moment tensor potential (MTP)[36] have also been employed to describe molten NaCl systems, yielding accurate predictions of properties such as solubility limits, standard reduction potentials, and transport coefficients[7]. Despite these advances, such potentials have been primarily developed and validated for single-component systems, particularly NaCl. Their transferability to multicomponent mixtures and chemically diverse environments remains insufficiently explored and has not yet been systematically demonstrated[7,37,38].

In this work, we develop and validate a moment tensor potential (MTP) for molten alkali chlorides, designed to accurately reproduce key liquid-state properties, including density, diffusion coefficients, heat capacity, thermal conductivity, and phase behavior over a broad temperature range. The MTP framework has demonstrated versatility across diverse material classes, including water and aqueous systems[39], inorganic compounds[40,41], metals and alloys[42,43], as well as complex ionic liquids and molten salts[37], highlighting its transferability across chemically distinct environments. For molten salts, MTPs are particularly valuable because they capture subtle local structural rearrangements and composition-dependent effects that are critical for describing liquid-state behavior. In this study, MTPs provide a robust and scalable framework for modeling molten alkali chlorides and their mixtures, efficiently representing many-body interactions while maintaining the accuracy necessary for comparison with experimental and first-principles data. Methodological details of the MTP formalism are available elsewhere[36,44]. By combining accuracy, transferability, and computational efficiency, the proposed model enables large-scale molecular dynamics simulations and provides a predictive framework for investigating multicomponent molten salt systems relevant to advanced clean energy technologies.

II. COMPUTATIONAL METHODS

A. Machine-learning potential

Among existing machine-learning interatomic potentials for materials modeling[35], we employ the moment tensor potential (MTP) framework[36,44]. The MTP formalism is based on a systematic tensor representation of local atomic environments, which enables an efficient and compact description of many-body interactions while preserving the essential symmetries of the atomic system, including translational, rotational, and permutational invariance. A key advantage of the MTP framework lies in its favorable balance between computational efficiency and predictive accuracy[45]. By expressing the potential energy as a linear combination of basis functions derived from systematically constructed moment tensor descriptors, MTPs achieve near first-principles accuracy at a computational cost slightly higher than classical semi-empirical potentials[45], yet significantly lower than density functional theory, making them suitable for large-scale simulations.

The formalism can be described mathematically as follows. The total energy of a system is computed as a sum of local atomic contributions:

$$E_{\text{Total}} = \sum_{i=1}^{N} E_i = \sum_{i=1}^{N} V_{\text{local}}(\zeta_i), \tag{1}$$

where $\zeta_i = (r_{ij}, \tau_i, \tau_j)$ encodes the relative positions r_{ij} and atomic types τ_i, τ_j. The local energy function V_{local} is evaluated within a cutoff radius R_c (here, 5.7 Å). Within this sphere, V_{local} is expanded in terms of basis functions B_β:

$$V_{\text{local}} = \sum_{\beta} c_\beta B_\beta, \tag{2}$$

allowing forces on atom k to be calculated via

$$F_k(x_q) = -\nabla E_{\text{Total}}(x_q) = -\sum_i \frac{\partial V_{\text{local}}(\zeta_i)}{\partial r_k(x_q)}. \quad (3)$$

The corresponding virial stress for a configuration of volume Ω is

$$\sigma_{ij}(x_q) = \frac{1}{2\Omega} \sum_{k\in\Omega} \sum_{l\in\Omega} (x_i^{(l)} - x_i^{(k)}) F_j^{(kl)}. \quad (4)$$

The basis functions B_β are constructed by contracting local tensor descriptors that encode both radial and angular distributions of neighboring atoms:

$$M_{\mu,\nu}(r_{ij}, \tau_i, \tau_j) = \sum_j f_\mu(|r_{ij}|, \tau_i, \tau_j) \underbrace{r_{ij} \otimes \cdots \otimes r_{ij}}_{\nu \text{ times}}. \quad (5)$$

Here, the radial functions f_μ are expanded in a set of radial basis functions $Q^{(\alpha)}$ with coefficients $c^{(\alpha)}_{\mu,\tau_i,\tau_j}$:

$$f_\mu(|r_{ij}|, \tau_i, \tau_j) = \sum_\alpha c^{(\alpha)}_{\mu,\tau_i,\tau_j} Q^{(\alpha)}(|r_{ij}|), \quad (6)$$

providing a flexible and accurate description of interatomic interactions. The full set of model parameters, $\Theta = \{c_\beta, c^{(\alpha)}_{\mu,\tau_i,\tau_j}\}$, is determined by minimizing a loss function that balances reproduction of reference energies, forces, and stresses.

B. Ab initio calculations

The reference database for training and validating the interatomic potentials was generated using density functional theory (DFT)[24] calculations carried out with the Quantum ESPRESSO package[46]. Ultrasoft pseudopotentials[47] were employed in conjunction with the generalized gradient approximation (GGA)[48], using the Perdew–Burke–Ernzerhof (PBE) functional[49] to describe the exchange–correlation energy. This setup provides a reliable balance between computational efficiency and accuracy for modeling ionic and molten systems (REF???). The kinetic energy cutoff for the plane-wave basis set was set to 900 eV, ensuring convergence of total energies, atomic forces, and stress tensors across the diverse configurations included in the reference database. This cutoff was determined through systematic convergence tests and is consistent with the requirements of the employed ultrasoft pseudopotentials. Given that the database contains atomic configurations with varying numbers of atoms, cell volumes, and densities—representing different thermodynamic conditions and structural motifs—a Monkhorst–Pack k-point mesh[50] was employed for Brillouin zone integration. The k-point density was scaled according to the simulation cell size to ensure consistent reciprocal-space sampling across all configurations. Detailed information on the specific k-point meshes used for each system is provided in the Supporting Information.

C. Active learning algorithms and MD simulations

We employ active learning to efficiently construct a reliable training database while minimizing the number of expensive *ab initio* calculations[51]. This approach automatically identifies atomic configurations that contribute most informatively to the potential, thereby avoiding redundant sampling of configurational space. In the MTP active learning scheme[51,52], an initial set of configurations is represented in the space spanned by the MTP basis functions. Initially, an active set is selected from the training set such that the modulus of the determinant of the corresponding matrix is maximized[51]. This determinant represents the volume occupied by the active set in configurational space. In other words, the selected configurations span diverse regions of configurational space. During molecular dynamics simulations, newly encountered configurations are evaluated against the current active set. The algorithm determines whether a configuration can increase the determinant of the active set by replacing an existing configuration. A configuration is included in the training set only if it increases the determinant of the active set matrix, indicating that it lies outside the region already well described by the current potential. In practice, extrapolation grades are embedded in the algorithm to classify each encountered configuration during the simulation as belonging to the interpolation regime, the extrapolative regime, or a risky extrapolation (unphysical configuration). Configurations are selected for training when the extrapolation grade is greater than 2 and less than 10. When the extrapolation grade exceeds 10, the simulation is terminated. The training set is then updated with new *ab initio* data, the potential is retrained, and the active set is updated. This iterative procedure—comprising initial training set generation, construction of the active set, on-the-fly identification of informative configurations, and incremental enrichment of the database—enables the efficient development of accurate and transferable MTP models while significantly reducing the computational cost associated with exhaustive *ab initio* sampling.

All MD simulations were performed using the LAMMPS package[53]. A time step of 1 fs was used throughout to ensure numerical stability and accurate integration of the equations of motion. Depending on the physical properties being investigated, simulations were conducted within the NVE, NVT, and NPT ensembles. The temperature was controlled using the Nosé–Hoover thermostat, while pressure was controlled using the Parrinello–Rahman barostat in the NPT ensemble, with the temperature damping parameter set to 100 fs and the pressure damping parameter set to 1000 fs, respectively.

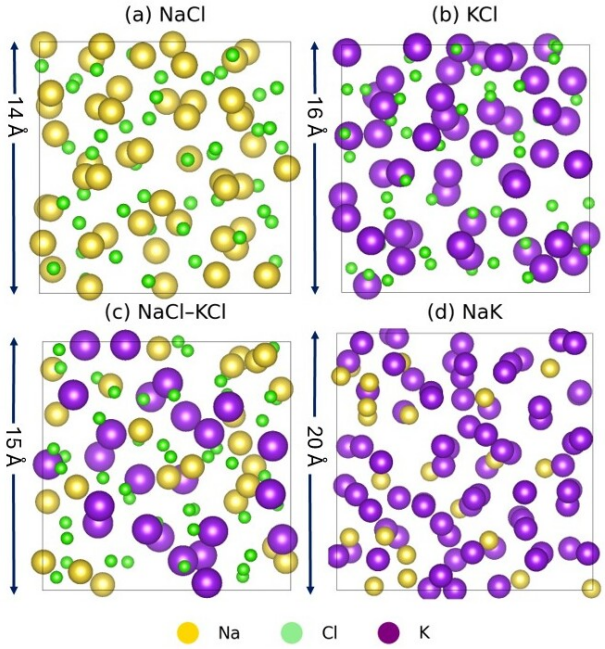

Fig. 1 Atomic structures of (a) molten NaCl, (b) molten KCl, (c) molten equimolar NaCl–KCl mixture, and (d) liquid metallic NaK alloy. Each system contains 96 atoms.

D. Dispersion energy-correction D3

The *ab initio* reference data used to train the interatomic potential were generated without dispersion corrections. To provide a physically consistent description of interatomic interactions, the D3 dispersion correction[54,55] was applied in molecular dynamics simulations. Coulombic forces dominate at short range, while long-range dispersion interactions contribute significantly to the cohesion of molten salts. Previous studies have shown that in the absence of van der Waals corrections, MD simulations of molten NaCl can exhibit unphysical expansion[56], highlighting the stabilizing role of dispersion forces. The D3 correction has been demonstrated to accurately reproduce both the microscopic structure and thermodynamic properties of molten NaCl[56], ensuring a reliable treatment of long-range interactions in combination with the machine-learned interatomic potential.

High-temperature molten salts such as NaCl and KCl exhibit pronounced short-range structural order. Neutron and high-energy X-ray diffraction measurements of molten NaCl have shown that structural correlations decay over a characteristic length of approximately 6.7 Å, corresponding to about 2.5 times the nearest-neighbor Na–Cl distance (≈2.7 Å)[15]. This indicates that the dominant structural order in the melt is governed by the first and second coordination shells, with only weak, screened correlations beyond this range. Motivated by this experimental observation, we employed reasonable cutoff for both the MTP potential and the D3 dispersion correction, ensuring that dispersion effects are included only within the physically relevant correlation length of the liquid. Thus, for the MTP potential, a cutoff of 7.5 Å was employed. For the D3 correction, a cutoff of 6 Å was used for the pairwise dispersion energy, and a cutoff of 6 Å was also used to evaluate coordination numbers for damping the dispersion interactions.

E. Simulation of melting point and phase diagram

To validate our MTP potential, we simulated the phase diagram of NaCl–KCl for compositions ranging from 10 % to 90 % NaCl mol, using pure NaCl and pure KCl as compositional endpoints. All simulations considered zero-pressure solid–liquid phase behavior. The Moving Interface (MI) method[57] was employed to determine the melting points of the pure salts, while the Compositional Moving Interface (CMI) method[58] was used for intermediate compositions. Simulations were performed in the NPT ensemble with a solid–liquid interface (Figure 2) at multiple temperatures.

For the pure end members, the interface advanced toward the solid or liquid depending on whether the temperature was below or above the melting point. The total energy was recorded over time, and the rate of energy change was obtained from a linear fit. The melting temperature was then identified as the temperature where this rate equals zero, consistent with the equilibrium criterion for single-component systems.

For the mixtures at intermediate NaCl compositions, the melting point was determined by monitoring the interface, the system energy, and the phase compositions[57]. The interface was considered stationary when the solid and liquid coexisted without net growth of either phase. Energy stabilization was confirmed when the total energy showed negligible change over time. Compositional equilibrium was achieved when net diffusion between the solid and liquid phases ceased, meaning the local compositions in each phase remained constant. In multicomponent systems, the solid and liquid generally have different equilibrium compositions, so the rate of energy change alone is insufficient to identify the melting point. By accounting for both interface stability and the cessation of net diffusion, we ensure that both phases have reached their equilibrium compositions, enabling an accurate determination of the melting temperature.

F. Free energy calculation

The miscibility gap constitutes a stringent thermodynamic test of whether an interatomic model accurately captures the energetics of mixing and phase stability in multicomponent systems. To assess the capability of our MTP potential to reproduce the full phase behavior of NaCl–KCl mixtures, we compute the miscibility gap from free energies obtained via thermodynamic integration[59]. The thermodynamic driving force for mixing is quantified by the Gibbs free energy of mixing. For a binary system of composition x, it is defined relative to the unmixed reference state as

$$\Delta G_{\mathrm{mix}}(x) = G_{\mathrm{hom}}(x) - [x\, G_A + (1-x)\, G_B] \qquad (7)$$

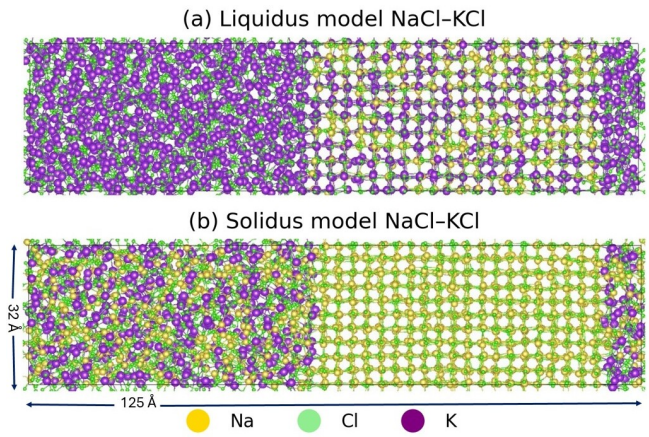

Fig. 2 Illustration of the atomic models used in the compositional moving interface method for evaluating melting temperatures in multicomponent mixtures. Each system contains 4000 atoms.

where $G_{\mathrm{hom}}(x)$ is the Gibbs free energy of the homogeneous mixture, and G_A and G_B are the Gibbs free energies of the pure components evaluated at the same temperature and pressure.

For the NaCl–KCl system, this expression becomes

$$\Delta G_{\mathrm{mix}}(x) = G_{\text{NaCl-KCl}}(x) - [x\, G_{\mathrm{NaCl}} + (1-x)\, G_{\mathrm{KCl}}] \tag{8}$$

The miscibility gap can also be identified by monitoring the Gibbs free energy of mixing at fixed compositions as a function of temperature. For a given composition x, the miscibility gap corresponds to the temperature at which

$$\Delta G_{\mathrm{mix}}(x,T) = G_{\text{NaCl-KCl}}(x,T) - [x\, G_{\mathrm{NaCl}}(T) + (1-x)\, G_{\mathrm{KCl}}(T)] = 0 \tag{9}$$

At temperatures above this point, the homogeneous mixture is thermodynamically stable, whereas below it, phase separation is favored. By computing ΔG_{mix} across a range of compositions, the full miscibility gap of the NaCl-KCl system can be determined. All mixture models with varying compositions were constructed in a two-step procedure. First, the endpoint compounds, NaCl and KCl, were relaxed using our MTP potentials to determine their respective equilibrium lattice parameters. The lattice parameter of a NaCl–KCl mixture with composition x was then estimated using Vegard's law:

$$a(x) = x\, a_{\mathrm{NaCl}} + (1-x)\, a_{\mathrm{KCl}} \tag{10}$$

where a_{NaCl} and a_{KCl} denote the lattice constants of the pure components. In the second step, the KCl lattice was constructed with lattice parameter $a(x)$, and a fraction x of the K sites were substituted with Na atoms to generate the desired mixture composition.

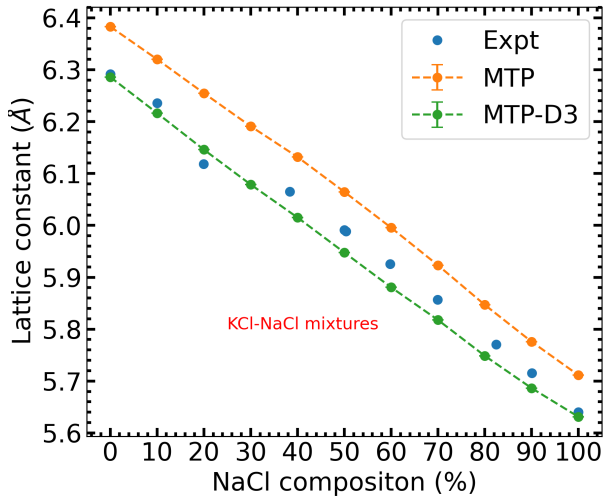

Fig. 3 Lattice parameters of NaCl–KCl mixtures as a function of NaCl composition (mol% x), predicted using MTP and MTP–D3, compared with experimental data.

III. Results and discussion

A. Bulk properties

The MTP predicts a bulk modulus of 30.7 GPa for NaCl and 16.6 GPa for KCl, compared to experimental values[60] of approximately 24 and 17 GPa, respectively. The MTP prediction for KCl is in close agreement with experiment, however the bulk modulus of NaCl is significantly overestimated. Such deviations are comparable to those reported for other machine-learning interatomic potentials and likely reflect limitations of the underlying DFT reference data, as well as the relative weighting of elastic deformations in the training set. In particular, the training database contains a very small fraction (less than 1 %) of strained crystalline configurations, which may limit the accuracy of the predicted elastic properties. Figure 3 shows the lattice parameters of NaCl, KCl, and their mixtures as a function of composition. As can be observed, the MTP model with the D3 correction yields lattice parameters in closer agreement with experiment[60,61] than the MTP model without dispersion. This demonstrates the importance of including dispersion interactions to accurately capture the equilibrium structural properties of alkali halide systems.

B. MTP predictions on the liquid-phase atomic structures

To characterize the performance of the MTP in predicting liquid structure, we computed liquid densities for NaCl, KCl, NaCl–KCl mixtures, and the NaK alloy, as shown in Figure 4. Three sets of densities are compared: those obtained using the MTP, MTP-D3, and experimental reference values[11,62,63]. Except for NaCl, densities predicted without dispersion corrections show significant deviations from experimental data, whereas inclusion of the D3 cor-

rection systematically improves agreement. For KCl, the uncorrected MTP density underestimates the experimental value by 18.36%, while the D3-corrected result reduces this error to 1.3%. For the NaCl–KCl mixture, the deviations are 10.83% without correction and 1.91% with D3, and for the metallic NaK alloy, 8.06% and 0.08%, respectively. In contrast, NaCl exhibits the opposite trend: the D3-corrected density underestimates experiment by 5.90%, whereas the uncorrected MTP result is closer, with a deviation of only 1.96%.

To further characterize the liquid structure, radial distribution functions (RDFs) were computed for three systems: NaCl, KCl, and the equimolar NaCl–KCl mixture. Four datasets were examined: MTP without dispersion correction, MTP with D3 correction (MTP-D3), DFT without dispersion correction, and DFT with D3 correction, as shown in Figure 5. Overall, both MTP and MTP-D3 reproduce the DFT liquid structures with high fidelity.

No significant structural differences are observed among the four datasets. This behavior is expected because the RDFs reflect averaged structural information. Moreover, the system volume was fixed for all RDF calculations, constraining the bonding environment and suppressing the artificial density expansion often observed in molten salts in the absence of D3 corrections[56]. As a result, dispersion effects are largely captured at the mean-field level—that is, as an averaged thermodynamic contribution—without substantially altering the local structure reflected in the RDFs.

C. Transport properties of liquid systems

For real-world applications, the transport properties of molten systems are of primary importance. Accordingly, we evaluate the predictive capability of the developed interaction potential against a set of key transport properties, including the self-diffusion coefficients, heat capacity, and thermal conductivity. Detailed descriptions of the molecular dynamics simulation protocols and the analysis methodologies employed to compute these properties are provided in the Supporting Information.

Figure 6 presents the thermal conductivity of NaCl, KCl, and the equimolar NaCl–KCl mixture along with available experimental data[64,65]. The MTP predictions include error bars representing statistical uncertainties. In all cases, the predicted values fall within the experimental ranges, demonstrating good overall agreement with experiment. Using the same interaction potential, we next evaluate the constant-pressure heat capacity of these molten systems, including the metallic NaK alloy. The corresponding results are shown in Figure 8 together with available experimental measurements[66? ,67]. The MTP predictions also include statistical error bars. For all four systems, the MTP results exhibit good agreement with experiment, despite minor differences between the simulation conditions and the experimental settings. In particular, for the equimolar NaCl–KCl mixture, the MTP prediction lies slightly below the experimental value but remains in close agreement, indicating reasonable results.

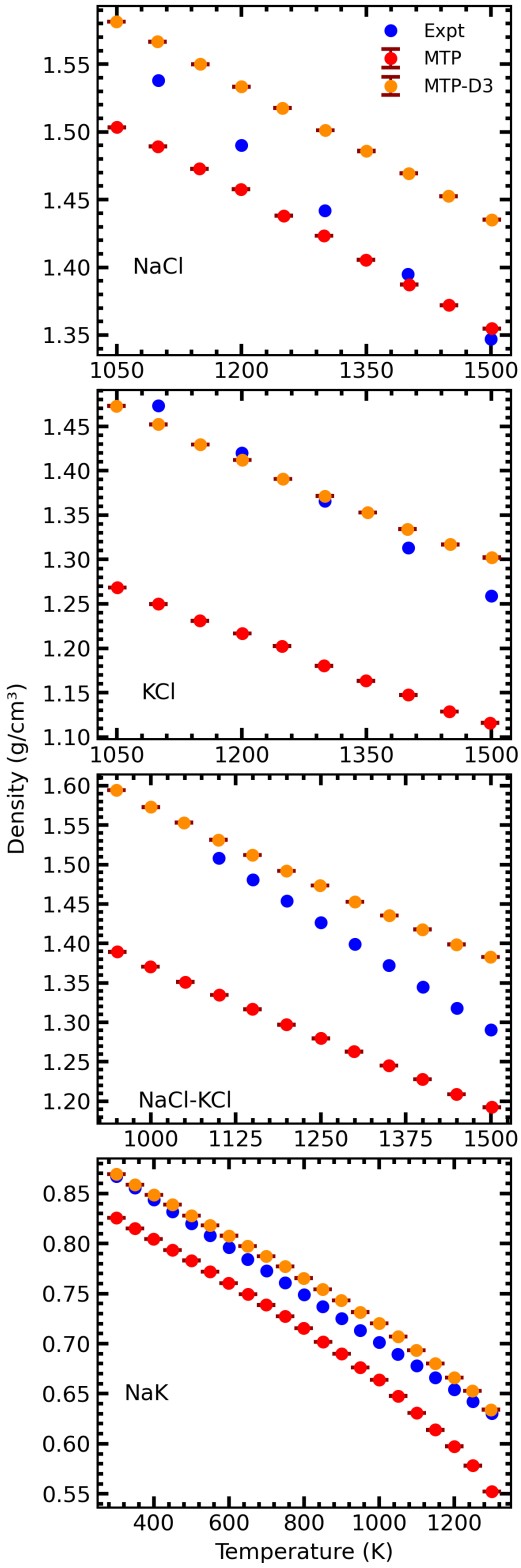

Fig. 4 Temperature-dependent densities of (a) NaCl, (b) KCl, (c) the equimolar NaCl–KCl mixture, and (d) the Na–K alloy, predicted using MTP and MTP–D3 and compared with experimental measurements. Error bars are small, indicating minimal variance.

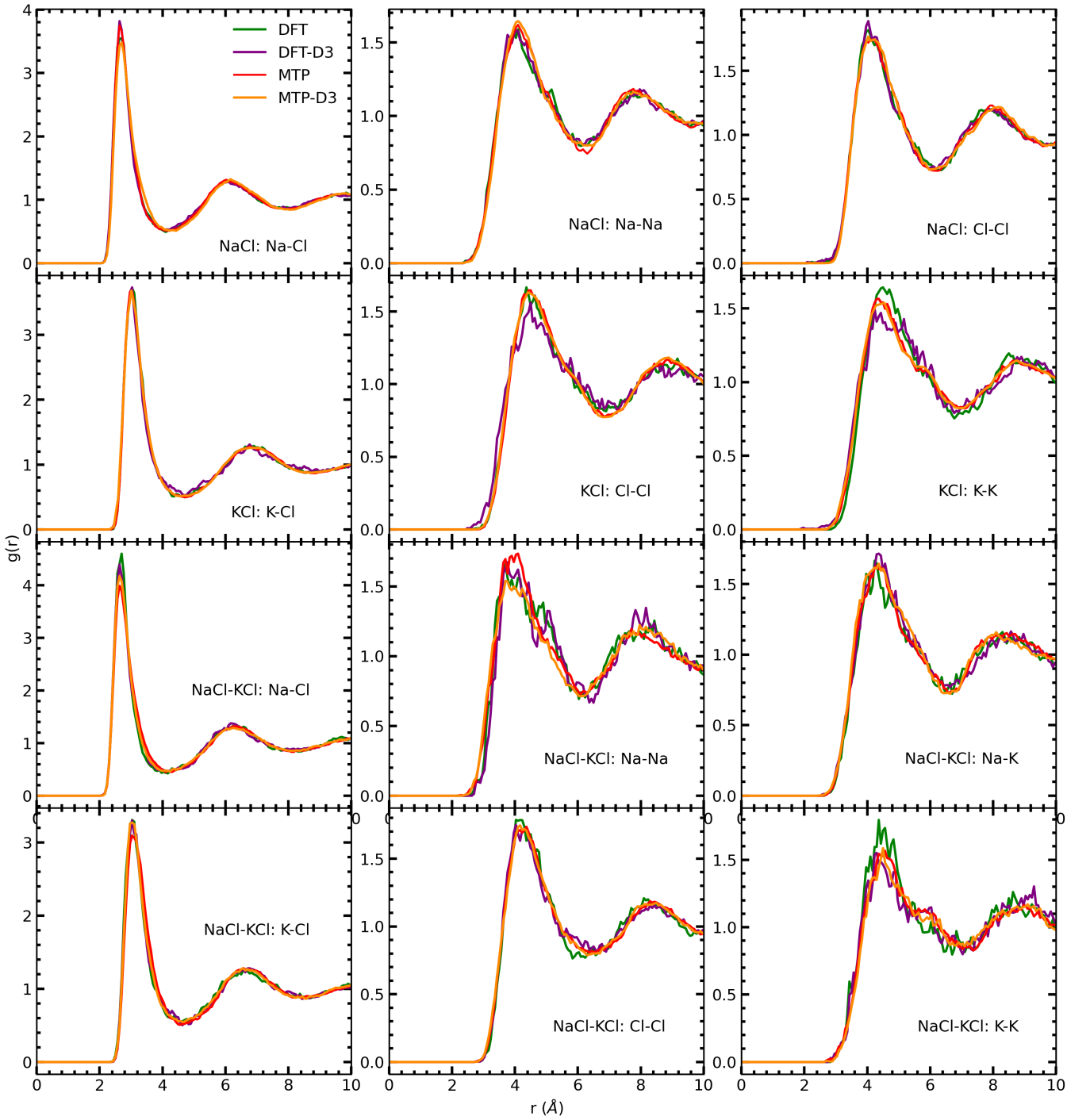

Fig. 5 Partial radial distribution functions of NaCl, KCl, and NaCl–KCl systems from 96-atom simulations. MTP and MTP–D3 predictions are compared with DFT and DFT–D3 data, with all four datasets superimposed to illustrate agreement. Simulation cell densities were set to experimental values at 1150 K (NaCl), 1100 K (KCl), and 1100 K (NaCl–KCl), and simulations were conducted at these temperatures.

This small discrepancy can be attributed primarily to differences in thermodynamic conditions: the simulations were performed at zero pressure with an exactly equimolar composition (50 mol% NaCl–50 mol% KCl), whereas the experimental data correspond to a slightly off-equimolar mixture (51 mol% NaCl–49 mol% KCl) measured under ambient conditions.

In addition, we measured the self-diffusion coefficients for NaCl, KCl, and the equimolar NaCl–KCl mixture. Figure 9 shows the MTP predictions together with avail-

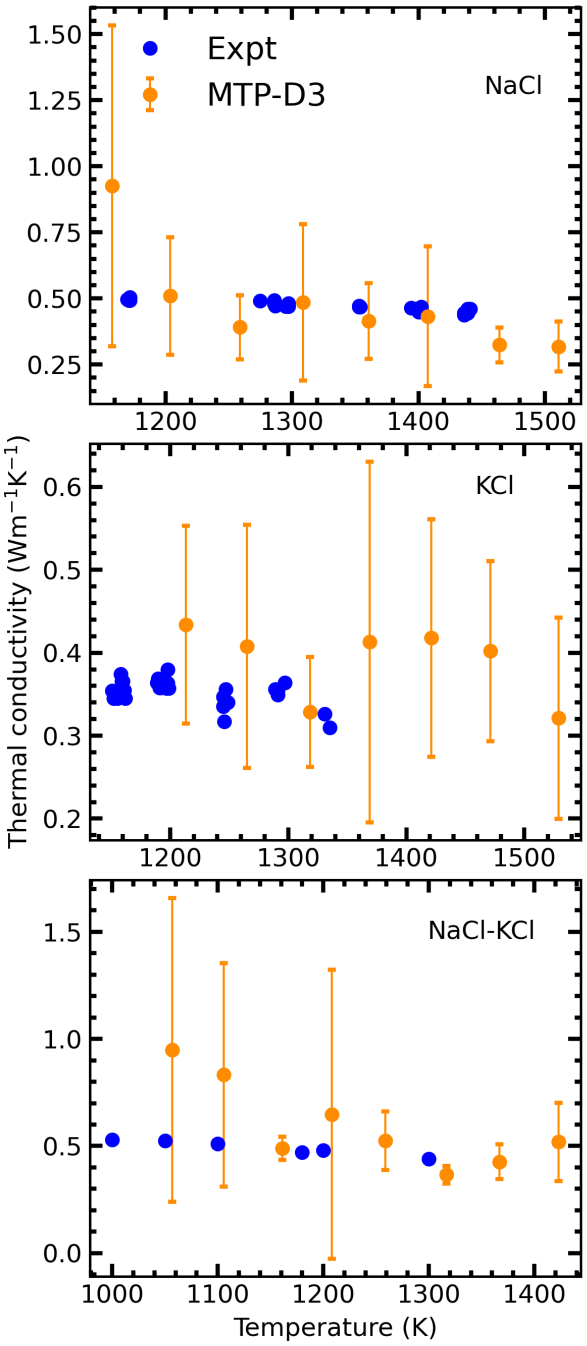

Fig. 6 Thermal conductivity of molten NaCl, KCl, and the equimolar NaCl–KCl mixture as predicted by MTP–D3, compared with experimental measurements. Error bars represent the standard deviation of the simulation data.

able experimental results[11]. For both Na^+ and Cl^- ions, the MTP results systematically underestimate the experimental diffusion coefficients across the entire temperature range, while correctly reproducing the monotonic increase with temperature. This behavior is consistently observed for all three systems considered in this work, indicating that the discrepancy is systematic rather than system-

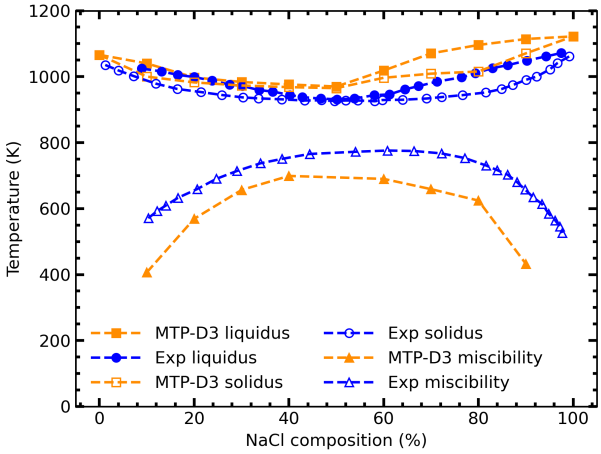

Fig. 7 Temperature–composition phase diagram of NaCl–KCl mixtures for compositions from 10% to 90% NaCl, with pure NaCl and KCl as reference endpoints. The liquidus and solidus lines, as well as the miscibility gap, predicted by MTP–D3, are compared with experimental data, highlighting the model's ability to capture phase stability and two-phase regions across the compositional range.

specific. The underestimation of the absolute diffusion values can be attributed primarily to finite simulation time and finite-size effects. In particular, the production runs were limited to 10 ns, which may not be sufficient to fully converge long-time diffusive behavior in highly viscous molten salts. Longer simulations are expected to yield higher diffusion coefficients and improve quantitative agreement with experiment. Nevertheless, the correct temperature dependence and consistent trends across all systems demonstrate that the MTP reliably captures the underlying transport physics.

D. Melting points and phase diagram

The predicted melting temperatures of the end members, KCl and NaCl, are 1004.3 and 1147.6 K, respectively, in good agreement with the experimental values of 1043 and 1073 K, corresponding to deviations of 3.7% and 6.9%. Such deviations are consistent with the expected accuracy of first-principles-based interatomic potentials, for which melting points are sensitive to the choice of pseudopotential, exchange–correlation functional, and dispersion corrections. The calculated solidus and liquidus lines for intermediate compositions between 10% and 90% NaCl are presented in Figure 7, together with the miscibility limits and corresponding experimental data for comparison. For the liquidus line, deviations from experiment range from 4.3% to 8.5%, with the largest discrepancy observed near 70% NaCl. The solidus line shows deviations between 2.5% and 7.6%, while the miscibility limits deviate by 3.4%–6.7%, both with the largest differences occurring around 50% NaCl. Overall, the predicted melting behavior and phase diagram demonstrate good agreement with experimental observations.The remaining discrepancies likely arise from limitations in the pseudopotential

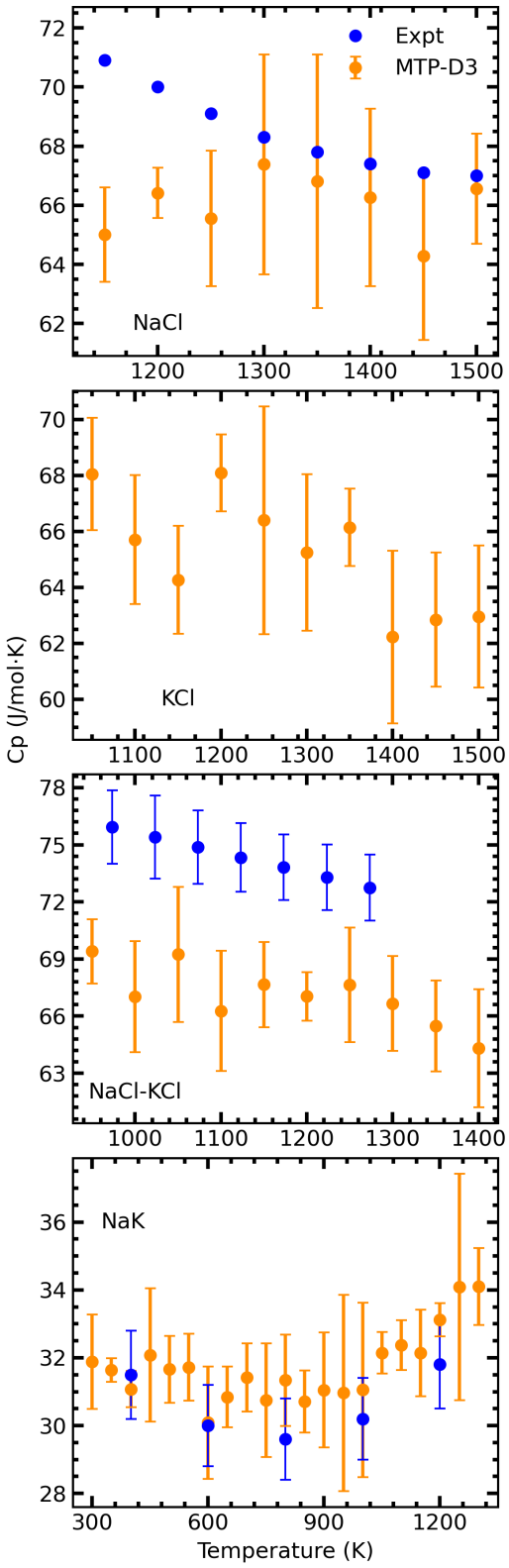

Fig. 8 Constant-pressure heat capacity of molten (a) NaCl, (b) KCl, (c) the equimolar NaCl–KCl mixture, and (d) the Na–K alloy as a function of temperature. MTP–D3 predictions with statistical uncertainties are compared with experimental data, demonstrating good agreement across all systems

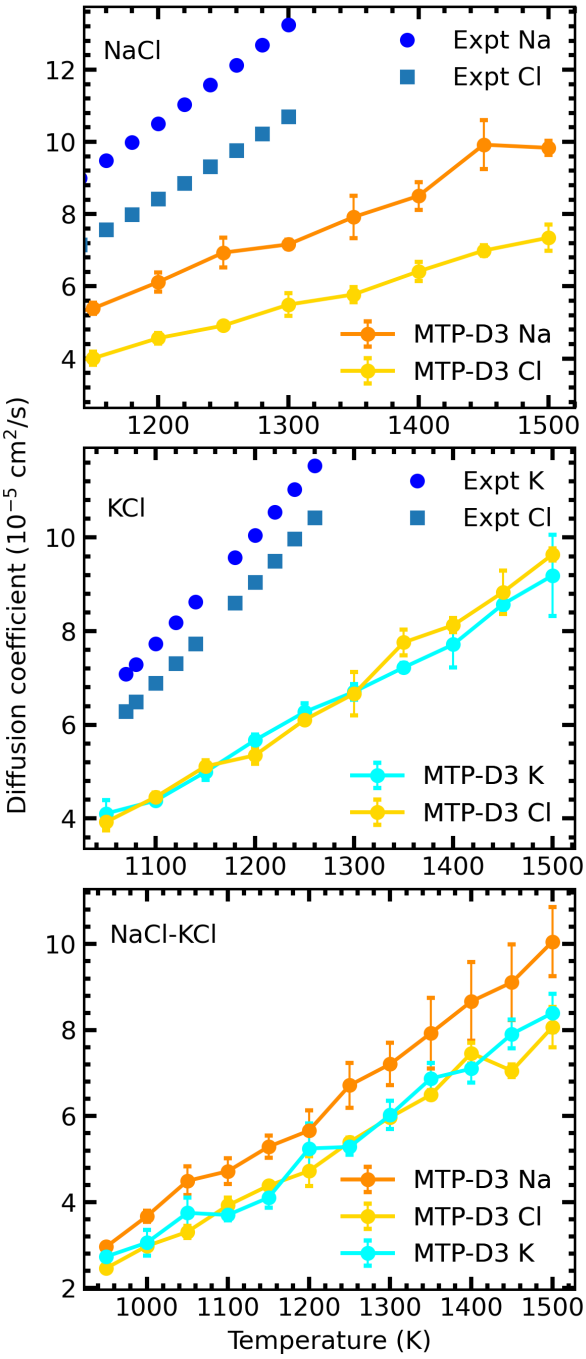

Fig. 9 Temperature-dependent self-diffusion coefficients of Na^+, Cl^-, and K^+ ions in molten (a) NaCl, (b) KCl, and (c) the equimolar NaCl–KCl mixture. MTP–D3 predictions, including statistical uncertainties (error bars), are compared with available experimental data to assess the accuracy of the model across different systems.

description of melting behavior and the relatively modest system size (4000 atoms) used for computational efficiency, compared with larger-scale simulations (>20,000 atoms) reported in the literature[58]. In addition, a systematic investigation of the D3 cutoff for each system and

composition would be necessary to identify the optimal value that yields results closest to experimental data.

E. Effect of D3 dispersion correction on all simulation results

To evaluate the influence of long-range dispersion interactions on our predictions, the D3 correction was applied during the MD simulations, while the DFT training dataset itself did not include D3. This section examines how the inclusion of D3 affects key properties such as lattice constants, thermodynamic properties of mixing (e.g., mixing enthalpy and phase stability), transport properties, and melting behavior. For consistency, the Becke–Johnson (BJ) damping form of D3 was employed with a fixed pairwise cutoff across all simulations.

Maintaining consistent dispersion settings ensures that observed variations in predicted properties arise from the inclusion of dispersion interactions rather than arbitrary changes in simulation parameters. Overall, incorporating D3 improves agreement with experimental data, particularly for properties sensitive to long-range interactions. Although a systematic exploration of alternative D3 functional forms and cutoff values was not performed in this study, the results clearly demonstrate the significant impact of dispersion corrections on the predictive capability of the MTP relative to the underlying DFT reference.

This effect is further illustrated by test calculations of melting temperatures using different D3 functional forms and cutoff parameters for NaCl and KCl. For NaCl, the use of the original D3 functional with a cutoff of 4.5 Å yielded a melting temperature of 1047.3 K, corresponding to a deviation of 2.4% from the experimental value of 1073 K. In contrast, employing the D3(BJ) damping scheme with a cutoff of 5.0 Å resulted in a melting temperature of 1067.1 K, in excellent agreement with experiment, with a deviation of only 0.55%. For KCl, the original D3 functional with cutoffs of 15 Å and 10 Å produced a melting temperature of 1065.2 K, deviating by approximately 2.2% from experimental data.

These results highlight the sensitivity of thermodynamic predictions to the choice of dispersion functional and cutoff parameters. Achieving closer quantitative agreement with experiments for a given system—such as NaCl, KCl, or NaCl–KCl mixtures—therefore requires careful selection and, ideally, systematic optimization of D3 settings. This becomes particularly important when varying the system type, phase (solid or liquid), or composition in multicomponent mixtures.

Conclusion

In this study, we developed and validated machine-learning interatomic potentials based on the moment tensor potential framework for molten NaCl, KCl, their mixtures, and the NaK system. By combining density functional theory reference data with an active learning strategy, we constructed reliable and transferable potentials capable of accurately describing liquid-phase behavior across diverse temperatures and compositions. The models successfully reproduce key thermophysical and structural properties, including density, heat capacity, radial distribution functions, self-diffusion coefficients, thermal conductivity, and phase behavior. Structural predictions agree closely with experimental and first-principles data, demonstrating the MTP framework's ability to capture local ordering and composition-dependent effects. Calculated melting temperatures and phase diagrams further confirm the model's capability to describe phase stability and transitions. The inclusion of D3 dispersion corrections prevents unphysical density expansion and highlights the importance of long-range interactions for realistic simulations of molten salts. Overall, this work establishes a robust and computationally efficient modeling framework that bridges classical force fields and first-principles methods, enabling large-scale simulations of molten alkali chlorides relevant to thermal energy storage, nuclear systems, and other high-temperature applications.

Future studies could systematically evaluate D3 corrections across properties, temperatures, and compositions, explore non-equimolar mixtures, identify optimal salt compositions, and compare D3 incorporation at the DFT level versus only in molecular dynamics, providing deeper insight into the most accurate and efficient modeling strategies for molten salts.

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Author contributions

L.K.B. initiated and coordinated the research project and provided the necessary computing resources. K.Z. performed all DFT calculations, trained the MTP, conducted molecular dynamics simulations, and carried out the analyses presented in this work. All authors—K.Z., H.S., E.T., C.M., and L.K.B.—contributed to writing the manuscript.

Competing interests

The authors declare no conflict of interest.

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